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Paleomagnetism of intrusive rocks in the Coast Range of Oregon: Microplate rotations in middle Tertiary time

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ABSTRACT

Paleomagnetic study of intrusive rocks in the Oregon Coast Range shows clockwise rotation of the range relative to North America of 51.5° ± 30.0° since late Eocene–early Oligocene time, 44.5° ± 17.0° since middle Oligocene time, and possibly 32.0° ± 26.5° since middle Miocene time. The simplest interpretation of paleomagnetic data currently available for the Coast Range indicates that it rotated steadily at a rate of about 1.5°/m.y. for approximately the period 50 to 20 m.y. B.P. Events since the middle Miocene are entirely unresolved.

INTRODUCTION

An impressive amount of paleomagnetic data now supports and documents Tertiary clockwise rotation of the Oregon Coast Range (Cox, 1957; Simpson and Cox, 1977; Simpson, 1977; Cox and Magill, 1977; Plumley and Beck, 1977). Rotation, measured relative to the interior of North America, may total as much as 75° since the early Eocene. Rotated rock units within the Oregon Coast Range occur as far south as the Klamath Mountains and as far north as the Tillamook Highlands, a distance of more than 350 km (Fig. 1). Paleocurrent directions in widely distributed turbidite sediments determined by Lovell (1969) and Snavely and others (1964) and measurement of paleomagnetic directions in the same rocks by Simpson (1977) show rather convincingly that most, if not all, of the Oregon Coast Range rotated as a single coherent block. The scale of this block rotation is such that its "decollement horizon" (level of detachment from Earth's interior) probably was the asthenosphere.

Work in western Washington and southern Vancouver Island carries the pattern of clockwise rotations northward across the Canadian border, but there the picture that is emerging is considerably more complex. Moderate clockwise rotations are found in southwestern Washington at about the latitude of Olympia (Globerman and Beck, 1979; Beck and Burr, 1979), but little or no displacement is seen in preliminary paleomagnetic results for Eocene rocks from either the Willapa Hills area, just north of the Columbia River in the southwestern extremity of Washington (Wells and Coe, 1979), or for small volcanic exposures near Port Townsend and Bremerton, east of the Olympic Mountains (M. E. Beck, Jr., and D. C. Engebretson, unpub. data). *Counterclockwise* rotation is found for paleomagnetic directions in the East Sooke Gabbro of southern

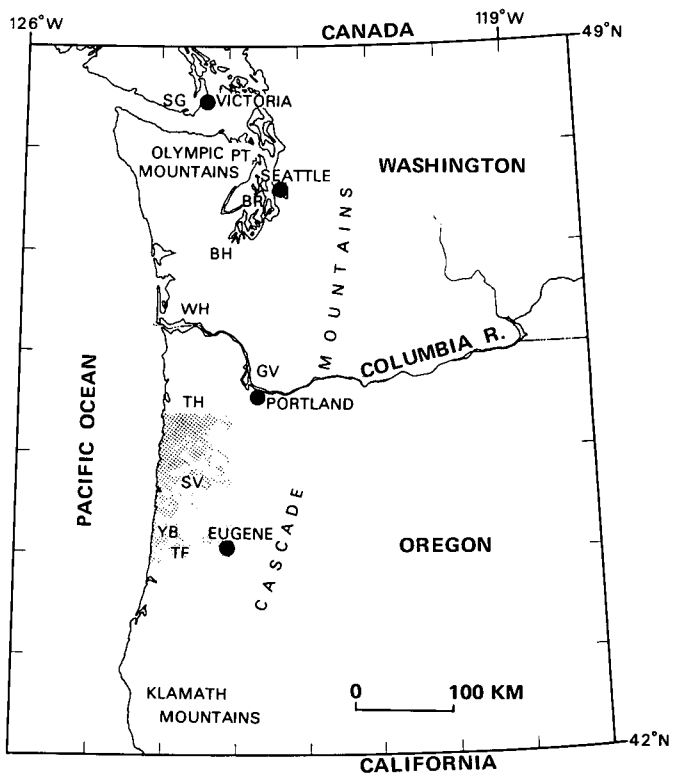


Figure 1. Map of Oregon Coast Range area. Stipple indicates area of Figure 2. SG = East Sooke Gabbro; PT = Port Townsend basalts; BR = Bremerton basalts; BH = Black Hills basalts; WH = Willapa Hills basalts; GV = Goble Volcanics; TH = Tillamook Highlands volcanic rocks; SV = Siletz River Volcanics; YB = Yachats basalts; TF = Tyee-Flournoy sedimentary rocks.

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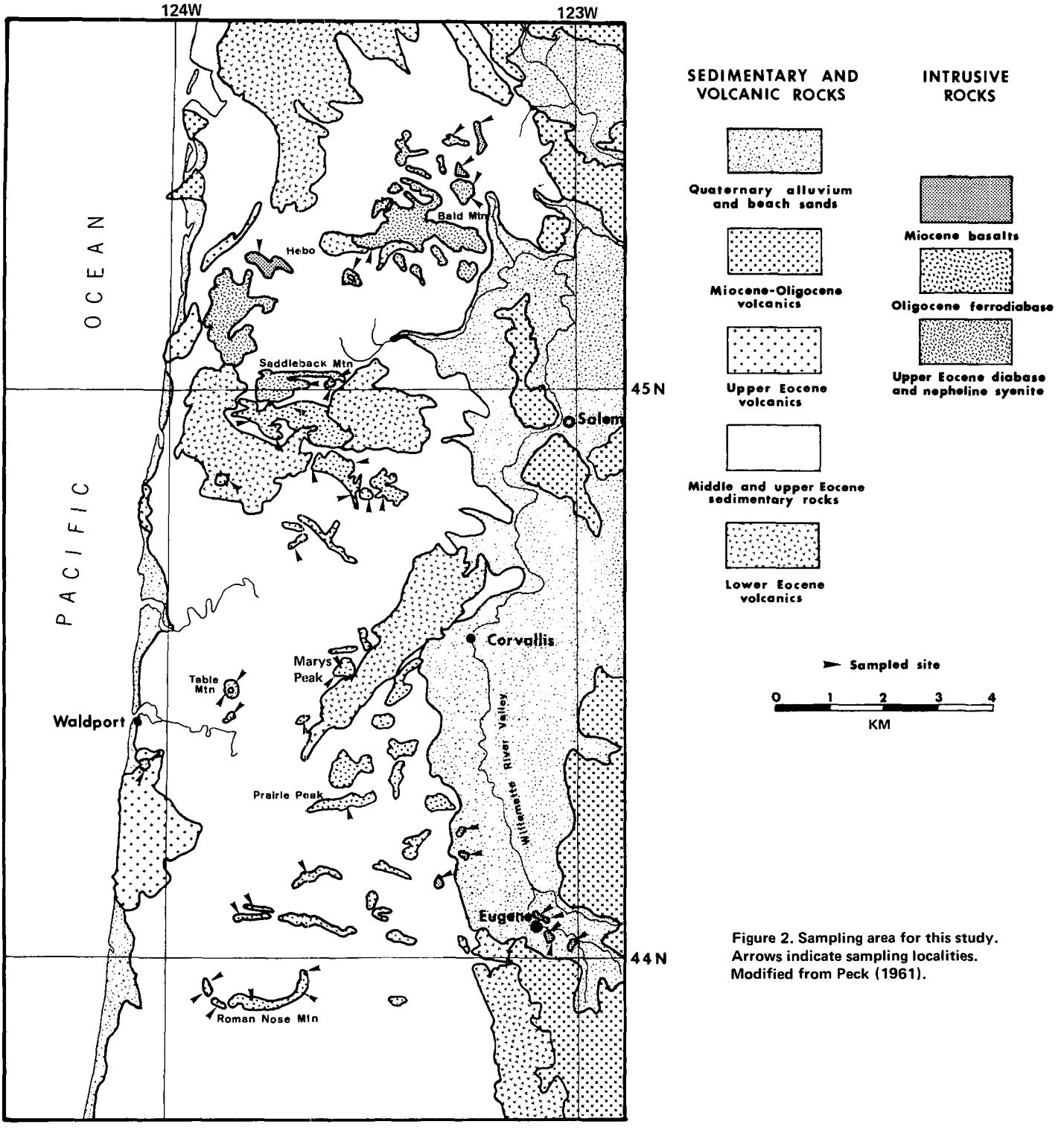


Figure 2. Sampling area for this study. Arrows indicate sampling localities. Modified from Peck (1961).

Vancouver Island (Symons, 1973), although this result may be affected by local tectonic disturbances associated with the Leech River fault. Thus, unlike the Oregon Coast Range, the Washington–Vancouver Island Tertiary exposures certainly have not rotated as a single rigid block. Moreover, the amount of rotation in Washington seems to have been less than in Oregon. Paleomagnetic results available so far suggest to us that at least two separate blocks are involved in Coast Range tectonics and that the present course of the Columbia River near its mouth marks an important tectonic boundary.

With the exception of the East Sooke Gabbro, which yields Oligocene K-Ar dates, all the paleomagnetic studies mentioned above are for units of Eocene age. Most of these are extrusive volcanic rocks, mainly basalts, and many are thought on geochemical grounds to be of oceanic derivation. The location of outcrops of Eocene rocks that have been sampled for paleomagnetic studies is shown in Figure 1; obviously, the study of Eocene tectonic rotations in the Coast Range is far advanced. Few attempts have been made to trace the chronology of rotation to its end, however, by studying progressively younger rocks in

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the Coast Range section. We report here on such an attempt, in which swarms of intrusive rocks were studied.

GEOLOGIC SETTING AND CHARACTERISTICS

The geologic setting of the Oregon Coast Range was conveniently summarized by Baldwin (1976). The early and early-middle Eocene basement of the Coast Range assemblage consists of pillow basalts of the lower Siletz River and lower Tillamook Volcanics, which pass upward into thick sequences of subaerial basalts. Overlying this volcanic basement are clastic marine sedimentary rocks, many of turbidity-current aspect. These in turn give way to Eocene subaerial basalt accumulations, such as the Yachats Basalt of the central Coast Range. The lower Eocene sequence has oceanic affinities; the volcanic piles, in particular, may represent oceanic islands built at a spreading center or as an intraplate seamount chain (see, for example, MacLeod and Snavely, 1973). Oligocene and Miocene sedimentary rocks also are dominantly marine, but in general they suggest deposition closer to shore, in shallower water. Presumably, this represents accretion of the Coast Range block to the craton during early and middle Tertiary time. Terrestrial volcanic rocks related to the Columbia River Basalt Group occur high in the Coast Range section.

Intrusive rocks of late Eocene, Oligocene, and Miocene age in the Coast Range were sampled for this study (Fig. 2).

Upper Eocene to lower Oligocene intrusive rocks consist of diabase sills of Saddleback Mountain (MacLeod, 1970), thin sills and dikes of porphyritic basalt petrographically correlated with the Yachats Basalt (Snavely and MacLeod, 1974), and widespread dikes and sills of alkalic rocks—specifically, camptonite and nepheline syenite (Snavely and others, 1976; MacLeod and Snavely, 1973). These rocks are dated on the basis of crosscutting relationships with fossil-bearing sedimentary rocks, supplemented by several K-Ar ages.

Iron-rich granophyric diabase sills of mid-Oligocene age are widespread throughout most of the central Coast Range, capping the high peaks and forming flat upland surfaces. Some of these sills are nearly 400 m thick. The distinctive chemistry (high silica and iron; low magnesia) and strong differentiation of these bodies make them readily recognizable. These intrusions also are dated on the basis of crosscutting relationships and radiometric age determinations.

Miocene intrusive rocks occur at several places along the flanks of the Coast Range. Thin sills of probable early Miocene age are exposed near the city of Eugene. Middle Miocene basaltic dikes and sills also are recognized on both flanks of the range; some of these are correlated petrographically with outcrops of extrusive rock (Snavely and others, 1973). Our sampling included Miocene flows as well as intrusive rock.

PREVIOUS PALEOMAGNETIC WORK

Clark (1969) studied the paleomagnetism of the Oligocene Marys Peak sill, near Corvallis. The paleomagnetic directions he measured are deflected sharply eastward from the expected Oligocene direction and have an anomalously shallow inclination. Both easterly declination and shallow inclination become more marked toward the center of the sill, so Clark concluded tentatively that the sill had cooled during an "excursion" of the geomagnetic field, when the virtual geomagnetic pole moved briefly into the Atlantic Ocean. Because of this and because of the short length of time presumably "sampled" by the sill while cooling through its magnetization temperature, Clark did not calculate a paleomagnetic pole from his results.

Miocene lava flows have been sampled at several points within the Oregon Coast Range; these results were summarized by Simpson and Cox (1977) and Choiniere and Swanson (1979). Other Miocene results bearing on the problem of tectonic rotation of the Coast Range were described by Kienle and others (1978). Within broad confidence limits these paleomagnetic data are compatible with little or no tectonic rotation of the Coast Range since about mid-Miocene time (15 m.y. B.P.).

PALEOMAGNETISM OF COAST RANGE INTRUSIVE ROCKS

In the course of three field seasons we collected more than 400 individually oriented samples at 60 separate sites. We aimed for at least 5 samples per site, but in a few small outcrop areas we collected as few as 3 (in some places an excess of caution and good drilling conditions led to the collection of as many as 16 samples). Samples were drilled and oriented in situ; standard paleomagnetic techniques were used. Magnetic orientations were checked by sun compass wherever possible, or by the back-azimuth technique when sun was not available. All samples were magnetically cleaned by alternating-field demagnetization, using a four-axis tumbler inside a low-field (less than 1,000 γ) space. Optimum demagnetization levels ranged from 50 to 250 Oe. For the most part, the samples were quite stable magnetically; most test specimens remained directionally stable to demagnetization levels of 500 Oe or more. Nevertheless, 5 sites were rejected because of excessive scatter (95% confidence circle greater than 15°). In addition, 5 sites were discarded because we could not be sure to which intrusive suite they belonged. Finally, multiple sites from single small intrusive bodies were combined on the assumption that they did not constitute independent measurements of the mid-Tertiary geomagnetic field; 30 such "duplicate" sites were combined into twelve composite groups. This left us with 32 reliable, independent, well-dated measurements of the geomagnetic field. Our results are summarized in Table 1; distributions of site-mean directions are shown in

TABLE 1. PALEOMAGNETIC DATA FOR MIDDLE TERTIARY INTRUSIVE ROCKS FROM THE OREGON COAST RANGE

Age	No. of sites	Declination (°)	Inclination (°)	K	α_{95}	Reference direction		Rotation (°)
						declination (°)	inclination (°)	
Late Eocene–early Oligocene	10	44.5	62.0	12.6	12.5	353.0	64.5	51.5 ± 30.0
Mid-Oligocene	14	39.0	56.0	20.0	8.4	354.5	63.5	44.5 ± 17.0
Miocene	8	29.5	67.5	26.8	9.6	357.5	63.5	32.0 ± 26.5

Note: K is precision parameter; α_{95} is radius of circle of 95% confidence. Reference direction calculated from polar-wandering curve for North America compiled by Irving (1977).

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Figure 3. Detailed site descriptions and statistics are given in Plumley (1980).

INTERPRETATION

From Figure 3, it appears that Coast Range rocks as young as middle Miocene have participated in the general tectonic rotation. Although scatter lessens their impact, all three plots show that all or nearly all site-mean directions (independent spot "readings" of the direction of the ancient geomagnetic field) lie well to the east of the expected direction. These reference directions are calculated from the known position of the pole, relative to the North American craton, as summarized by the polar-wandering curve of Irving (1977). As such, they represent the direction of magnetization each suite of rocks should display had they not been moved since they became magnetized. All three clusters are rotated eastward (clockwise); the amount of rotation increases with the age of the rock. Because observed and expected *inclination* are nearly identical, the Coast Range apparently has not moved significantly northward or southward, relative to North America, since late Eocene time. Rotations and their 95% confidence limits are given in the last column of Table 1. The sense of rotation in the Coast Range intrusive rocks exactly matches the pattern of paleomagnetic discordance reported earlier by Simpson and Cox (1977).

Figure 4, modified from Simpson and Cox (1977), summarizes Oregon Coast Range rotation data that have been published. Apparently, clockwise rotation at a rate of roughly 1.5°/m.y. persisted from early Eocene to early Miocene time, a span of some 30 m.y. At the beginning of Miocene time nearly half of the total Coast Range rotation remained to be accomplished.

For post–early Miocene time, Figure 4 illustrates two possibilities. The simpler is that rotation continued at a steady rate virtually until the present day. This possibility is not consistent with Simpson and Cox's (1977) Miocene data (MF in Fig. 4), and it does not take into account the conclusions of Kienle and others (1978). It should be noted that the sampling sites of Kienle and others (1978) lie northeast of an extrapolation of the Brothers fault zone (Lawrence, 1976) or the Oregon-Nevada lineament (Stewart and others, 1975), northwest through the Portland Hills fault and the mouth of the Columbia River to the sea. As we stated above, this line may mark an important tectonic boundary. Kienle and others' (1978) sampling sites therefore may lie adjacent to the Washington Coast Range block, not the Oregon Coast Range, and these two regions may have had different tectonic histories. However, if the Simpson and Cox (1977) MF data are taken at face value (or the Kienle results are applicable to the Oregon Coast Range), an interesting second possibility emerges. If Oregon Coast Range rotation was complete by middle Miocene time, then rotation velocities must have increased substantially during the early Miocene. After swinging sedately into place for 30 m.y., the Coast Range "door" (in this model) slammed shut with a bang! Only investigation of the paleomagnetism of middle Miocene and younger Oregon Coast Range rocks can definitely differentiate between these alternative models. However, the early Miocene was an important time of tectonic transition in the Cordillera (Coney, 1978), and if the Coast Range slammed shut as a result of that transition, its arrival in place coincided closely with an outbreak

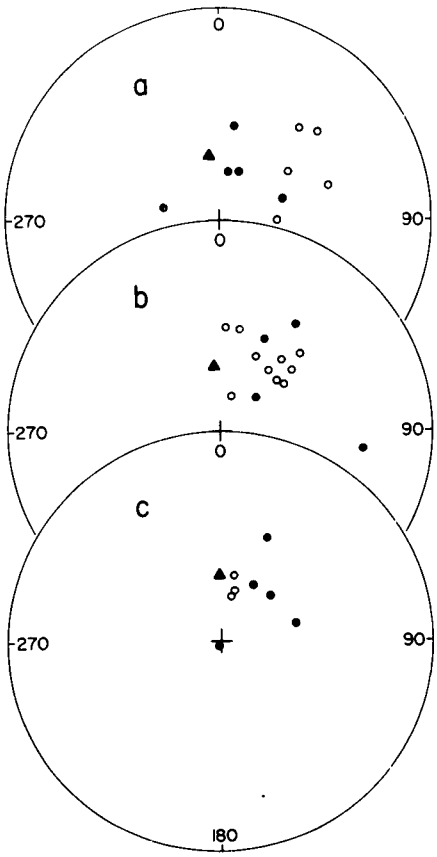


Figure 3. Site-mean directions of remanent magnetization. (a) Eocene intrusions; (b) Oligocene intrusions; (c) Miocene intrusions and lava flows. Triangle = expected direction of remanent magnetization, calculated from appropriate reference pole of Irving (1977). Large dots = normal magnetic polarity; circles = reversed magnetic polarity inverted through center of projection.

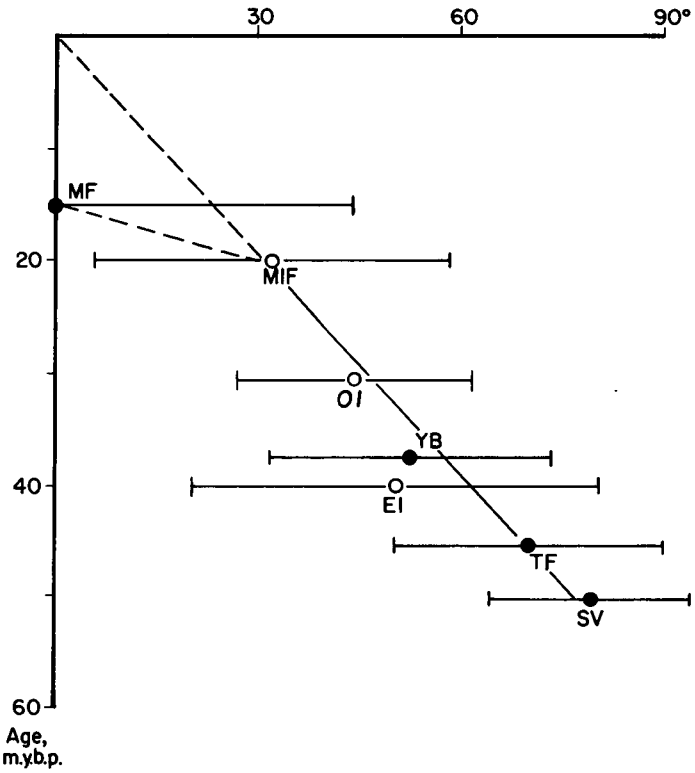


Figure 4. Diagram showing rotation of various Oregon Coast Range geologic units versus age. Large dots indicate data from Simpson and Cox (1977); circles indicate data of this study. Error bars give 95% confidence limits on rotation. SV, TF, YB as defined for Figure 1. EI = Eocene intrusions; OI = Oligocene intrusions; MIF = Miocene intrusions and flows; MF = Miocene flows. EI may include material as young as early Oligocene (N. S. MacLeod, 1980, personal commun.).

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of tremendous volcanic activity on the Columbia Plateau to the east (Donald A. Swanson, 1980, personal commun.). Initiation of extensional faulting and basaltic volcanism in the Basin and Range province also occurred at this time (Cole and Armentrout, 1979). We tend to see these events as causally related.

Qualifications

It will not have escaped the reader's attention that our interpretation of post-Miocene events in the Oregon Coast Range depends heavily on proper dating of our "Miocene" intrusive rocks. Likewise, the "swinging door" hypothesis for Miocene tectonics is wholly dependent on the reliability of paleomagnetic evidence showing that rotation of the Oregon Coast Range was complete by late Miocene time. Both points seemed to us reasonably secure, but both now appear to be vulnerable. Unpublished K-Ar dating of supposed "Miocene" rocks in the Coast Range, brought to our attention by N. S. MacLeod (1980, written commun.), shows a great deal of scatter, with some ages as old as late Eocene! A preliminary account of this work is given by Lux and Sutter (1978). To our knowledge, none of the supposed Miocene units we sampled have been dated by isotopic methods. Also, continued investigation of upper Miocene lava flows in the Columbia River Plateau and the Oregon-Washington Coast Range by R. Bentley, J. Magill, R. Simpson, and R. Wells has reopened the possibility of post–late Miocene rotation (R. Simpson, 1980, personal commun.). Thus, the matter is wholly unresolved. Rather than wait for a final solution of the Miocene question, we have elected to present our data as they now stand. The reality of our post-Oligocene rotation is unshaken by these new developments, but our Miocene data and interpretations should be eyed with suspicion.

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